Ryvath Mattey

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Education

University of Georgia Athens, GA

B.S. Computer Science, Minor in Microbiology *Aug. 2023 – May 2027*

- **Relevant Coursework:** Data Structures, Algorithms, Software Development, Computer Architecture, Databases
- **Certifications:** AWS Certified Cloud Practitioner, HIPAA Certified

Technical Skills

Languages: Java, Python, Go, TypeScript, JavaScript, SQL (PostgreSQL), C/C++, HTML/CSS
Frameworks & Libraries: Spring Boot, React, FastAPI, gRPC, PyTorch, Scikit-Learn, HAPI FHIR, Claude API
Infrastructure: Kafka, Redis, DynamoDB, Docker, Kubernetes, Terraform, AWS (EKS, RDS, Lambda, Cognito)
Tools & Testing: Git, GitHub Actions, Prometheus, Grafana, OpenTelemetry, Jaeger, Testcontainers, pytest, k6, Linux

Projects

Chartwise: AI Clinical Documentation Pipeline | *Spring Boot, Kafka, Go* github.com/Rybro63/chartwise

- Built an event-driven pipeline that drafts SOAP notes from visit transcripts with the Claude API, queues them for clinician review, and files approved notes to a FHIR server (synthetic Synthea data only)
- Used a transactional outbox and idempotent Kafka consumers so no encounter is lost or processed twice; killing the worker mid-stream across 200 encounters left 0 lost and 0 duplicated notes
- Built a Go gRPC gateway with Redis token-bucket rate limiting, jittered retries, and a circuit breaker; in a simulated 60-second LLM outage the breaker opened in 8s and the backlog drained with 0 failed drafts
- Load-tested with k6 against a 30s drafting SLO: 3 workers kept 100% of 1,029 drafts within it (fake LLM); a live Claude run exposed head-of-line blocking, and per-partition sliding commits cut p50 from 26.4s to 18.7s
- Wrote a medication safety check that caught 100% of 500 injected drug and allergy errors with 0 false positives on 400 clean drafts; encrypted PHI at rest and made the audit log append-only
- Deployed with GitHub Actions CI, Kubernetes (kind, and AWS EKS via Terraform), and OpenTelemetry tracing

batchserve: Dynamic Batching Inference Server | *Go, Concurrency* github.com/Rybro63/batchserve

- Built a request-batching inference server in Go (standard library only) that amortizes a fixed per-call backend cost across batched requests, raising goodput 12.8× (23 → 299 req/s) and eliminating timeouts at 300 req/s
- Implemented bounded-queue admission control, deadline dropping, and size-or-time flushing with distinct statuses for "never attempted" (503) vs. "backend failed" (500); validated with 44 tests under Go's race detector
- Showed queue depth dictates failure mode: a 1024-slot queue timed out 3,271 requests at 23 req/s, while a 16-slot queue shed load in microseconds and held 125 req/s at stable p99 – 5.4× more useful work

AI Symptom Checker (Full-Stack) | *React, FastAPI, AWS Lambda, DynamoDB, Cognito* live demo

- Built and deployed a multi-user AI triage app: React on Vercel, a serverless FastAPI backend on AWS Lambda, and the Claude API for assessments across 4 triage levels, with Cognito JWT auth scoping all data to each user
- Added a 0.35 confidence threshold that never withholds emergency signals; tested with 14 pytest tests

Work Experience

Undergraduate Researcher – VIPR AI/ML and Robotics Aug. 2024 – May 2025

The University of Georgia *Athens, GA*

- Developed Python scripts integrating robotic hardware with REST APIs and PyTorch, enabling real-time monitoring and adaptive decision-making in farm environments
- Designed an autonomous robotic workflow using YOLOv8 for floor-egg detection, reducing manual labor
- Benchmarked YOLOv5–YOLOv12 across five model sizes, achieving 98.4% detection accuracy (mAP50) with YOLOv12x while cutting training time 88% using YOLOv8n
- Automated RGB–depth alignment and dataset preprocessing in Python across 1,000+ poultry frames

Medical Scribe Intern Aug. 2023 – May 2024

Prime Health Medical Center *Buford, GA*

- Managed 30–40 patient records per shift with near-zero errors in QTC and Optum LHI EHR systems under HIPAA